

Optimal Log-Likelihood Tests for Distinguishing Generative Models under Relative Entropy Constraints

How to choose the detector model when the true source is out of reach

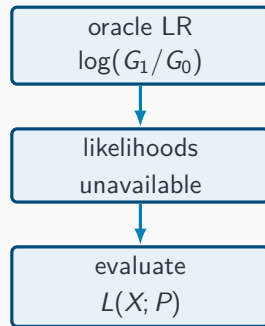
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Testing When Model Likelihoods Are Inaccessible

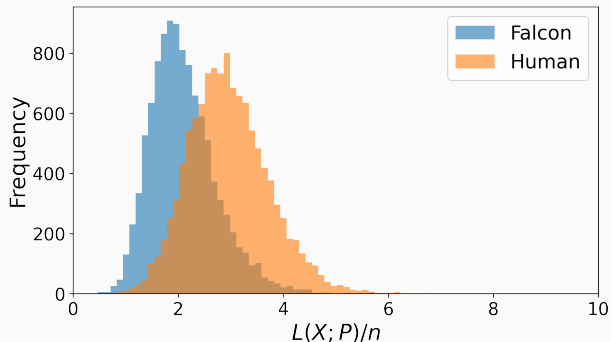
- Was text X generated by G_0 , or by G_1 ?
- **Oracle test:** threshold the likelihood ratio $\log \frac{G_1(x)}{G_0(x)}$.
- **Problem:** we can often *sample*, not evaluate likelihoods (closed source, compute heavy, inaccessible).
- **Practical test:** evaluate logarithmic loss $L(X; P) = \log \frac{1}{P(X)}$ under a third detector model P (aka perplexity test).



Goal: which detector P maximizes detection power?
selected by an optimality criterion, not learned inside the test

Example: Log-Loss from Two Sources

Wikipedia sentences from two authors: **Falcon-7B** vs **Human**, under a third model $P = \mathbf{Phi-2}$.



Related detectors: GLTR (Gehrmann et al., 2019), HowKGPT (Vasilatos et al., 2023), Binoculars (Hans et al., 2024).

The goal is maximum separation

Prior work addresses **adjacent problems** but not this **detector selection optimality question**:

- **Unconstrained optimal detection** - Neyman-Pearson; Lehmann-Romano (2006)
When both models are known, thresholding the likelihood ratio $\frac{G_1(X)}{G_0(X)}$ is optimal.
- **Mismatched divergence** - Unnikrishnan, Huang, Meyn, Surana & Veeravalli (2011)
Studies hypothesis testing when the ideal KL-based statistic is replaced by a mismatched one.
- **M-projection / reverse I-projection** - Amari & Nagaoka (2000)
Studies KL-based projection geometry over probability distributions; related to our objective, which is linear in $\log p$.

Problem Setup

Object: X is the data sequence to attribute.

Hypothesis test:

$$H_0 : X \sim G_0 \quad \text{vs.} \quad H_1 : X \sim G_1$$

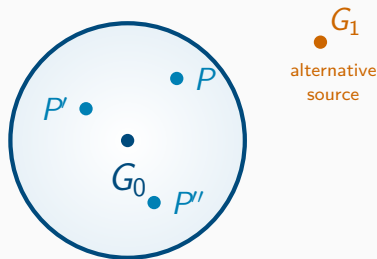
Test statistic: $L(X; P) := \log \frac{1}{P(X)}$

Modeling assumption: P lies near G_0 (shared tokenizer / architecture / training):

$$P \in \mathcal{P}_\epsilon(G_0) = \{P' : D(G_0 \| P') \leq \epsilon\}$$

Constrained optimality: among detectors in this ball, we search for the one with maximum detection power.

$$D(G_0 \| P) \leq \epsilon$$



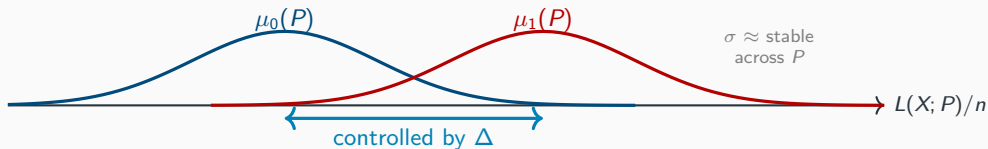
Result 1: Detector Power is Governed by Δ

Assumptions: under G_0 and G_1 , normalized log-loss is unimodal with variance stable across detectors.

Power is monotone in the separation between the two log-loss distributions:

$$\mu_1(P) - \mu_0(P) = \underbrace{H(G_1) - H(G_0)}_{\text{fixed}} + \underbrace{\Delta(G_1, G_0; P)}_{\text{depends on } P}$$

$$\Delta(G_1, G_0; P) := D(G_1 \| P) - D(G_0 \| P)$$

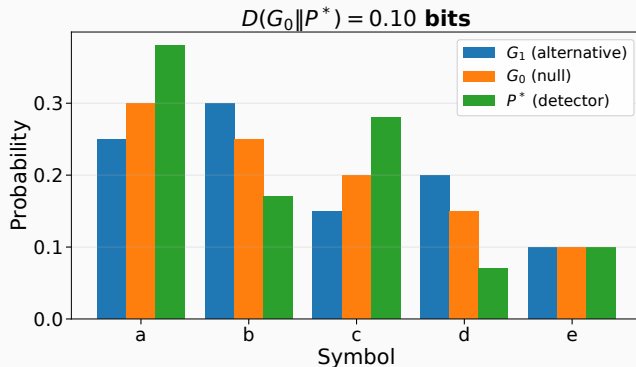


Theorem 1: maximizing Δ maximizes power at fixed Type-I level.

Result 2: Choosing the Optimal Detector P^*

$$\max_{P \in \mathcal{P}_\epsilon(G_0)} \Delta(G_1, G_0; P) \implies P^*(x) = G_0(x) - \gamma(G_1(x) - G_0(x))$$

Theorem 2: P^* moves linearly away from G_0 , opposite to G_1 , with $D(G_0 \| P^*) = \epsilon$.



Result 3: When Self-Detection is Locally Maximin-Optimal

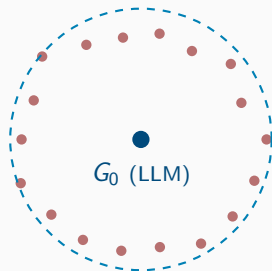
Single G_1 : move away from G_0 .

Family of alternatives $\mathcal{G}_1$: optimize the worst case.

$$\Delta^*(\mathcal{G}_1, G_0) = \sup_{P \in \mathcal{P}_\epsilon(G_0)} \inf_{G \in \mathcal{G}_1} \Delta(G, G_0; P)$$

Modeling assumption: LLM G_0 can be viewed as a learned convex mixture of human-style sources:

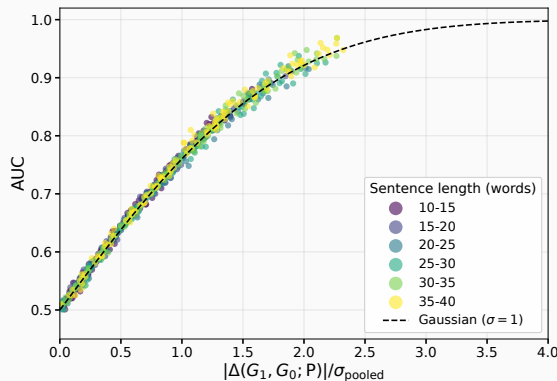
$$G_0 \approx \sum_j \pi_j H_j \quad \pi_j \geq 0, \sum_j \pi_j = 1$$



alternative sources
around G_0

If alternatives isotropically surround G_0 :
 $P = G_0$ is locally maximin optimal

Empirical Check: Normalized Separation Predicts AUC



$$x := \frac{|\hat{\mu}_1(P) - \hat{\mu}_0(P)|}{\hat{\sigma}_{\text{pooled}}}$$

$$\text{AUC} \approx \Phi(x/\sqrt{2})$$

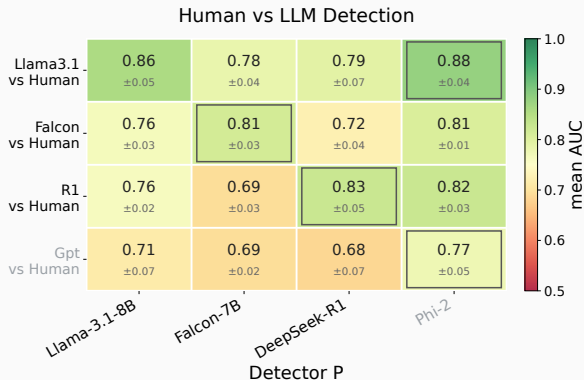
Experiment setup: 275,000 sentences, 3 domains, 5 authors, 4 detector models

Empirical AUC follows the Gaussian curve: [Pearson \$r = 0.99\$](#) .

Supports Result 1: Gaussian-like log-loss behavior and stable variance.

Max/min standard-deviation ratio across detectors is $1.2\text{--}1.5\times$.

LLM vs. Human: Support for the Maximin Choice $P = G_0$



Matched detector $P = G_0$:

- Best in 4/9 cases, second-best in 5/9
- Mean AUC **0.831** vs. 0.626 for LLM-vs-LLM

Consistent with the maximin model if LLM G_0 is surrounded by human sources.

1. When candidate-model likelihoods are unavailable, detect with a third model P .
2. The power-maximizing detector maximizes $\Delta(G_1, G_0; P) = D(G_1 \| P) - D(G_0 \| P)$.
3. For composite alternatives isotropically surrounding G_0 , self-detection is locally the maximin optimal choice.

Choose the detector. Maximize the gap.

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Extended version & proofs

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How Sub-Optimal Is $P = G_0$?

Small- ϵ expansion of the optimal gap:

$$\Delta(G_1, G_0; P^*) = D(G_1 \| G_0) + \sqrt{2\epsilon D_{\chi^2}(G_1 \| G_0)} + O(\epsilon).$$

Deviating from G_0 can raise the gap by $\approx \sqrt{2\epsilon D_{\chi^2}(G_1 \| G_0)}$, quantifying the sub-optimality of $P = G_0$ against a **simple** alternative.

Where this comes from: the $\gamma \Leftrightarrow \epsilon$ relation. The constraint $D(G_0 \| P^*) = \epsilon$ fixes the step size γ . It has no closed form in general, but for small ϵ

$$\epsilon \approx \frac{\gamma^2}{2} D_{\chi^2}(G_1 \| G_0) \quad \Longleftrightarrow \quad \gamma \approx \sqrt{\frac{2\epsilon}{D_{\chi^2}(G_1 \| G_0)}} \quad (\gamma \propto \sqrt{\epsilon}),$$

which substituted into the gap produces the $\sqrt{2\epsilon D_{\chi^2}(G_1 \| G_0)}$ term above.

Here $D_{\chi^2}(G_1 \| G_0) = \sum_x (G_1(x) - G_0(x))^2 / G_0(x)$; ϵ in nats (1/ln 2 factor if measured in bits).

The Maximin Mechanism

For a composite family $\mathcal{G}_1$, the local maximin gap is set by a least-favorable prior π^* :

$$\Delta^*(\mathcal{G}_1, G_0) = \mathbb{E}_{G \sim \pi^*}[D(G \| G_0)] + \sqrt{2\epsilon D_{\chi^2}(\bar{G}_{\pi^*} \| G_0)} + O(\epsilon),$$

$$\bar{G}_{\pi^*}(x) = \mathbb{E}_{G \sim \pi^*}[G(x)].$$

If a non-convex alternative family isotropically surrounds G_0 , this mechanism yields
Corollary 5.1: $P = G_0$ is locally maximin.

Anisotropy Index and Isotropic Optimality

For a prior $\pi^\#$ on $\mathcal{G}_1$ with $G_0 = \bar{G}_{\pi^\#} = \mathbb{E}_{G \sim \pi^\#}[G]$:

$$\bar{D}_{\pi^\#} = \mathbb{E}_{G \sim \pi^\#} D(G \| G_0), \quad r_{\pi^\#} = \sup_{G \in \text{supp}(\pi^\#)} |D(G \| G_0) - \bar{D}_{\pi^\#}|.$$

$r_{\pi^\#} = 0$ iff all sources are equidistant from G_0 in relative entropy (the mixture is isotropic); it grows with the unevenness of those distances.

For any alternative $G \in \mathcal{G}_1$, the self-detector $P = G_0$ obeys

$$\Delta(G, G_0; G_0) = D(G \| G_0) \geq \Delta^*(\mathcal{G}_1, G_0) - r_{\pi^\#} + O(\epsilon),$$

so its worst-case sub-optimality is at most $r_{\pi^\#}$. When $r_{\pi^\#} = 0$,

$$\inf_{G \in \mathcal{G}_1} D(G \| G_0) = \Delta^*(\mathcal{G}_1, G_0) + O(\epsilon),$$

i.e. $P = G_0$ is locally maximin optimal.

Experimental Setup

Datasets (3 domains, matched topics)

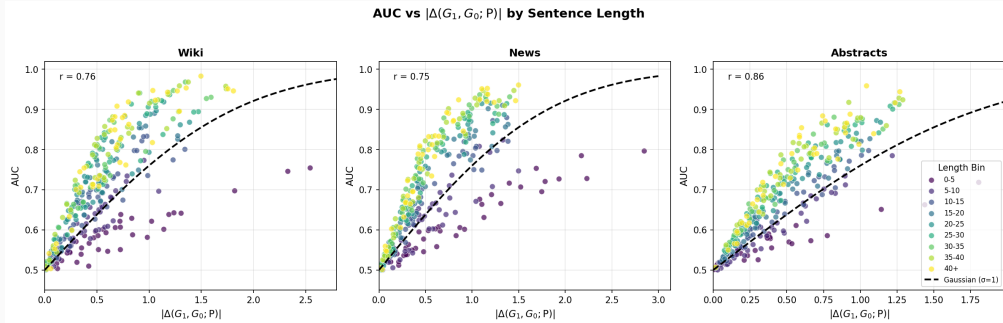
- Wikipedia, News, Scientific abstracts
- $\sim 4,500$ docs, $\sim 275,000$ sentences
- Sentence length: 10–40 words
- Scored without preceding context

Authors / Detectors

- Authors: Llama-3.1, Falcon, GPT-3.5, DeepSeek-R1, Human
- Detectors P : Llama-3.1, Falcon, Phi-2, DeepSeek-R1
- Some coincide $\Rightarrow$ self-detection ($P = G_0$)

GPT-3.5 model inaccessible for evaluation, so self-detection isn't possible.

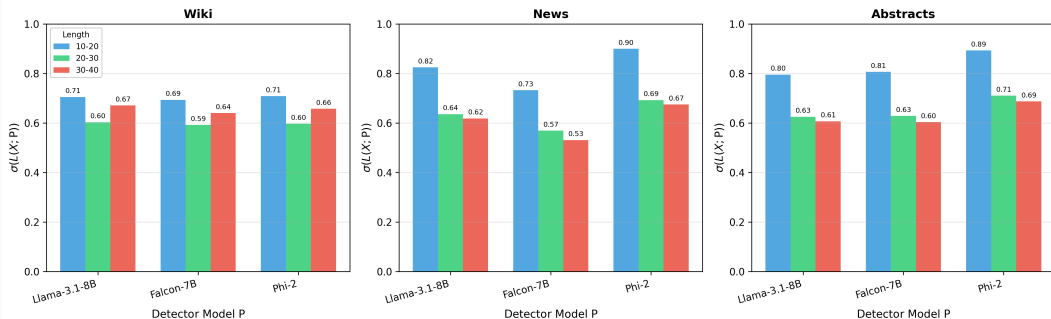
Why Normalize?



Raw log-loss gaps are confounded by length. Dividing by $\hat{\sigma}_{\text{pooled}}$ collapses length bins onto the Gaussian AUC curve used in the main evidence slide.

Variance Stability by Length

Standard Deviation of Log-Perplexity $\sigma(L(X; P))$ by Detector



A3 view: standard deviation $\sigma(L(X; P))$ by detector, split by sentence length. *Within each length bin*, σ is nearly identical across Llama-3.1, Falcon, and Phi-2 (across-detector ratio $\approx 1.0\text{--}1.3\times$), supporting A3; the larger spread within a dataset is driven by length, not by P . DeepSeek-R1 is excluded as an outlier (its chat template inflates σ).

What “Best in 4/9” Means

- **The 9 cases:** 3 domains $\times$ 3 matched LLM authors (Llama-3.1, Falcon, DeepSeek-R1).
- **GPT is excluded:** it has no corresponding open detector.
- **Phi-2 is detector-only:** it can win a heatmap cell, but is not part of the matched-detector count.
- **The claim:** among matched-detector cases, $P = G_0$ is best in 4/9 and second-best in the remaining 5/9.

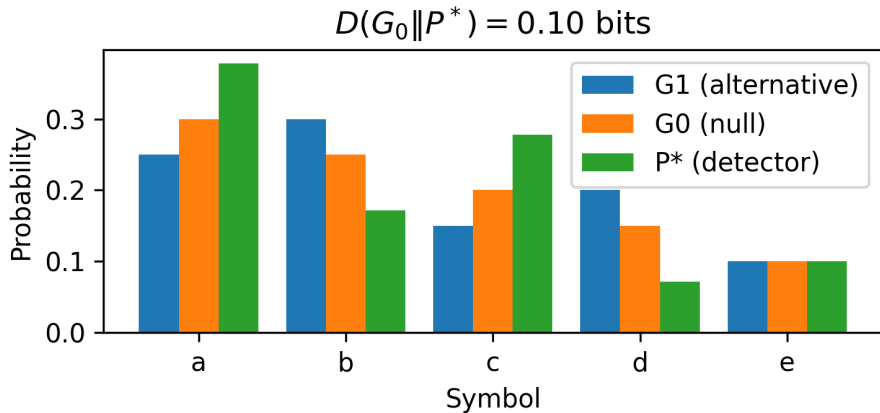
Matched-Detector AUC Numbers

AUC when detector matches source ($P = G_0$)

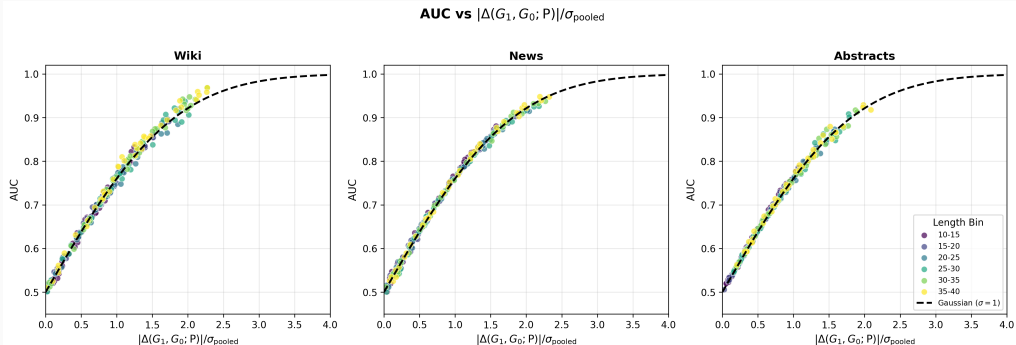
Dataset	Human vs. G_0	G_0 vs. G_1	Gap
Wiki	0.853 ± 0.006	0.659 ± 0.065	+0.194
News	0.854 ± 0.047	0.626 ± 0.094	+0.229
Abstracts	0.787 ± 0.018	0.592 ± 0.013	+0.195
Mean	0.831	0.626	+0.206

Human-vs-LLM self-detection is substantially easier than LLM-vs-LLM matched detection in this experiment.

Backup: Original P^* Example (paper figure)



Backup: Original AUC vs Normalized Separation (paper figure)



Backup: Original Human-vs-LLM Heatmap (paper figure)

